

Item	Material	Vol. frac	Density (kg/m3)	Mass (g/cell)	Mass (kg/kWh)	Formula / source
Positive electrode active material	NMC811	0.58	3262	9.715	0.988	$L_{\text{pos}} \times A \times \text{AMVF} \times d$ (parameter set)
Positive electrode binder	PVDF	0.015	1780	0.137	0.014	literature default density (no parameter in set)
Positive electrode conductive additive	carbon black	0.005	2200	0.056	0.006	literature default density (no parameter in set)
Negative electrode active material	graphite	0.56	1657	5.718	0.581	$L_{\text{neg}} \times A \times \text{AMVF} \times d$ (parameter set)
Negative electrode binder	SBR/CM C	0.02	1300	0.160	0.016	literature default density (no parameter in set)
Separator	polyolefin class	0.449999999996	397	0.165	0.017	$L_{\text{sep}} \times A \times (1 - \text{porosity}) \times d$ (parameter set)
Positive current collector	Al	1.0	2700	4.437	0.451	$L_{\text{al}} \times A \times d$ (parameter set)
Negative current collector	Cu	1.0	8960	11.042	1.123	$L_{\text{cu}} \times A \times d$ (parameter set)
Electrolyte	EC/EMC + LiPF6 class		1200	6.180	0.629	pore volume $\times 1.2 \text{ g/cm}^3$ (literature); excluded from contract mass
Enclosure / tabs	Not modeled			Not modeled	Not modeled	outside pure-simulation boundary
Component sum (excl. electrolyte)				31.431	3.196	sum of rows 1-8
Contract mass (calc-energy caliber)				31.616	3.215	cell/r4_v8_energy.json mass_kg
Total incl. electrolyte				37.611	3.825	contract mass + electrolyte fill